



0–5 min → Introduction & Hook

Ask:

“In VR, you can see objects and hear sounds—but why doesn’t it feel fully real?”

Explain:

Because **touch is missing**.

VR becomes truly immersive when users can:

- Feel objects
- Sense pressure
- Experience resistance

This is called **haptic interaction**.

Role of Touch in VR

Importance of Touch

Definition:

Haptic interaction refers to the use of touch-based feedback to communicate with the user in a virtual environment.

Why Touch is Important

Touch helps in:

- Realistic interaction
 - Object manipulation
 - Spatial awareness
 - User immersion
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Human Sensory System in VR

VR uses three major senses:

- Vision (eyes)
- Audio (ears)
- Touch (skin/muscles)

Without touch:

- Interaction feels artificial
 - No physical confirmation
-

Examples:

- Feeling vibration when pressing button
 - Feeling resistance when grabbing object
 - Feeling impact in VR games
-

Key Insight:

Touch confirms that an action has occurred.

Types of Haptic Feedback

Haptic Feedback Types

1. Vibrotactile Feedback

Double Vibration

Take your games to the next level



Definition:

Uses vibration to simulate touch sensations.

Examples:

- Controller vibration
- Phone vibration
- Game feedback

Advantages:

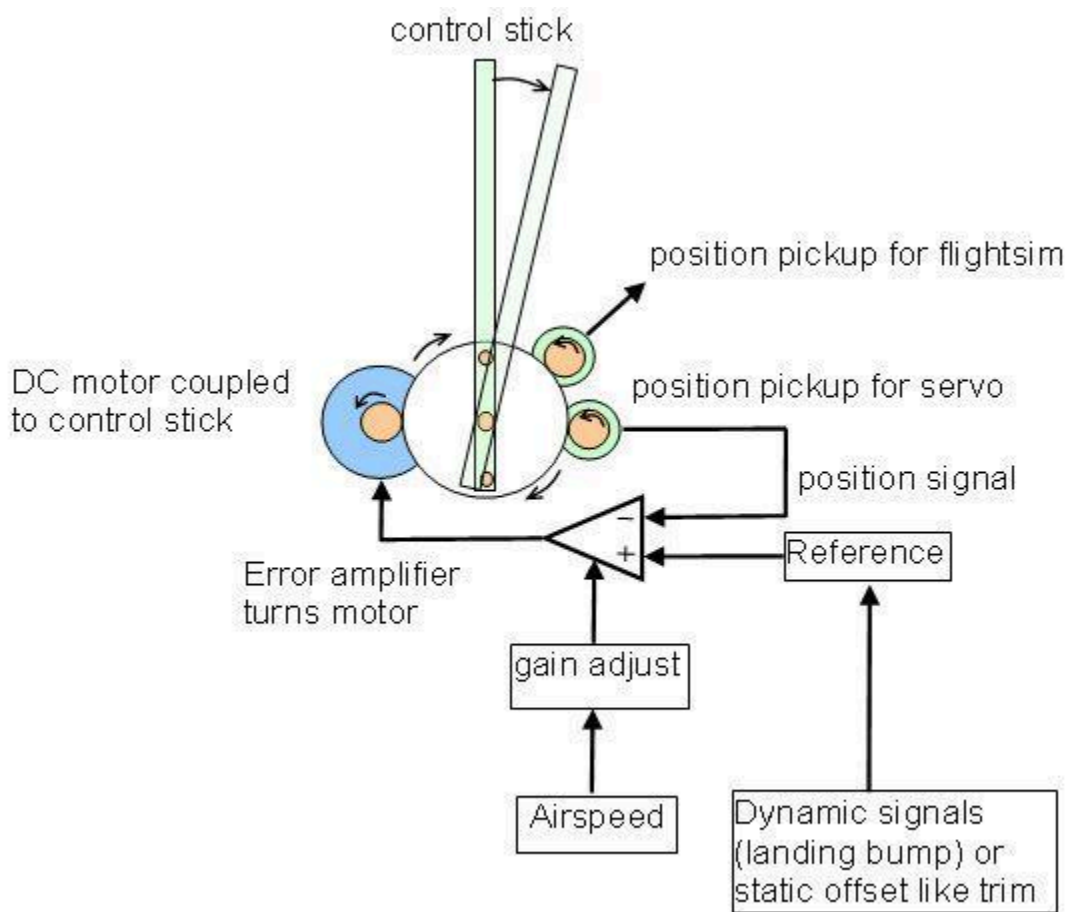
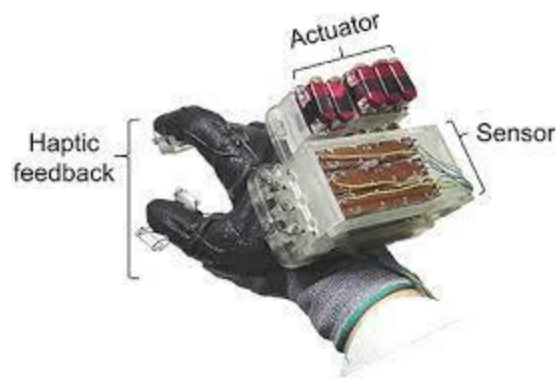
- Simple
- Low cost
- Widely used

Limitation:

- Cannot simulate force or texture accurately
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2. Force Feedback





Definition:

Provides resistance or force to simulate physical interaction.

Examples:

- Steering wheel resistance

- Haptic gloves
- Surgical simulators

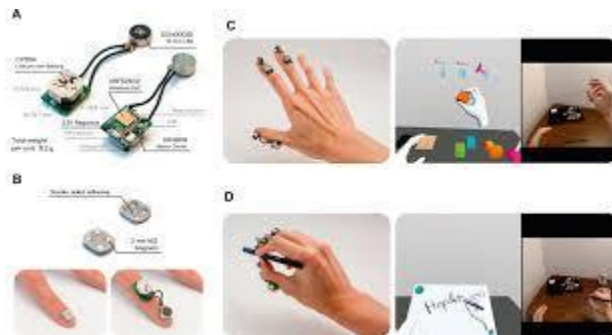
Advantages:

- High realism
- Accurate interaction

Limitation:

- Expensive
 - Complex hardware
-

3. Tactile Feedback



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Definition:

Simulates surface properties like texture, pressure, or shape.

Examples:

- Feeling rough vs smooth surface

- Pressure feedback on fingertips
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Advantages:

- Detailed touch sensation
 - Useful in medical and training VR
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Limitation:

- Complex to implement
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Haptics + VR Interaction

Integration with VR System

Interaction flow:

```
User Action
↓
Input Detection
↓
System Processing
↓
Haptic Feedback
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Example:

User grabs object:

- System detects grip
 - Object attaches
 - Controller vibrates
 - User feels feedback
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Real Applications:

- VR gaming
 - Medical simulation
 - Industrial training
 - Rehabilitation
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Important Concept:

Haptics closes the interaction loop by providing physical feedback.

Real VR Example

VR surgery simulation:

- Visual → see organ
- Audio → hear tool
- Haptic → feel resistance

This creates full immersion.

Recap & Questions

Key Points:

- Haptic interaction adds touch to VR
 - Enhances realism and immersion
 - Types: vibrotactile, force, tactile
 - Used in training, gaming, and simulation
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Quick Questions:

1. What is haptic interaction?
2. Why is touch important in VR?
3. What are the types of Haptic feedback?
4. What is vibrotactile feedback?

5. Difference between force and tactile feedback?